WORKSHEET 3.2

Curves That Are Not Functions: Implicit Differentiation

Unit 3: Composite, Implicit and Inverse Functions

Key ideas

- Differentiate both sides with respect to x, treating y as a function of x.
- Every term containing y produces a factor $\frac{dy}{dx}$: $\frac{d}{dx}\left[y^3\right] = 3y^2\frac{dy}{dx}$.
- A term such as xy needs the **product rule**: $y + x\frac{dy}{dx}$.
- Collect the $\frac{dy}{dx}$ terms, factor, and divide. Substitute a point only *after* differentiating.

Worked examples

Example 1 · A circle

Find the slope of $x^2 + y^2 = 29$ at the point $(5, 2)$.

Solution. $2x + 2y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{x}{y}$.

The slope is $-\frac{5}{2}$.

Example 2 · A product term

Find $\frac{dy}{dx}$ for $x^3 + xy + y^2 = 7$, and its value at $(1, 2)$.

Solution. $3x^2 + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{3x^2 + y}{x + 2y}$.

At $(1, 2)$: $-\frac{5}{5} = -1$.

Example 3 · A trigonometric relation

Find $\frac{dy}{dx}$ if $\sin y = x^2$.

Solution. $\cos y\,\frac{dy}{dx} = 2x$.

$\frac{dy}{dx} = \frac{2x}{\cos y}$.

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Practice

CORE

1. Find $\frac{dy}{dx}$ for $x^2 + y^2 = 20$.

2. Find $\frac{dy}{dx}$ for $y^3 + y = x$.

3. Find the slope of $xy = 18$ at the point $(3, 6)$.

4. Find the slope of $4x^2 + y^2 = 8$ at the point $(1, 2)$.

EXAM LEVEL

5. Find $\frac{dy}{dx}$ for $x^2y + y = 10$, and its value at $(2, 2)$.

6. Find the slope of $e^y + xy = e$ at the point $(0, 1)$.

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7. Write an equation of the line tangent to $x^2 - xy - y^2 = 1$ at the point $(2, 1)$.

8. Find the points on the curve $x^2 + y^2 - 4x = 0$ at which the tangent line is horizontal.

Question 9 · Free-response practice

Consider the curve $x^2 + 3y^2 = 12$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y.

(b) Find the coordinates of the points on the curve at which the slope of the tangent line is 1.

(c) Find the coordinates of the points on the curve at which the tangent line is vertical.

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Worked solutions

1. $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

2. $(3y^2+1)\dfrac{dy}{dx} = 1$, so $\dfrac{dy}{dx} = \dfrac{1}{3y^2+1}$.

3. $y + x\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{y}{x} = -2$.

4. $8x + 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{4x}{y} = -2$.

5. $2xy + x^2\dfrac{dy}{dx} + \dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{2xy}{x^2+1}$; at $(2,2)$ this is $-\dfrac{8}{5}$.

6. $e^y\dfrac{dy}{dx} + y + x\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{y}{e^y+x} = -\dfrac{1}{e}$.

7. $2x - y - x\dfrac{dy}{dx} - 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = \dfrac{2x-y}{x+2y} = \dfrac{3}{4}$: $y - 1 = \dfrac{3}{4}(x-2)$.

8. $2x + 2y\dfrac{dy}{dx} - 4 = 0$, so $\dfrac{dy}{dx} = \dfrac{2-x}{y}$, zero at $x = 2$. Then $y^2 = 4$: the points $(2,2)$ and $(2,-2)$.

9. (a) $2x + 6y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{x}{3y}$.

   (b) $-\dfrac{x}{3y} = 1$ gives $x = -3y$. Then $9y^2 + 3y^2 = 12$, so $y = \pm 1$: the points $(-3,1)$ and $(3,-1)$.

   (c) The denominator $3y$ is zero when $y = 0$, and then $x^2 = 12$: the points $\left(2\sqrt{3}, 0\right)$ and $\left(-2\sqrt{3}, 0\right)$.

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